Attacks and Defense

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https://community.trustcloud.ai/kbuPFACeFReXReB/uploads/2023/09/Cybersecurity-Cheetos-Meme.jpeg

Background

- Web development a very popular profession!
- Involves creating and maintaining websites or web applications
  - Lot goes behind the scenes, as we saw!
- Can introduce various attack vectors
  - Can be exploited by malicious players

Attacks

- A good characterization of attacks?
  - Server-side attacks
  - Client-side attacks
- Each targets different components of the web architecture
  - Client-side: User's browser and device
  - Server side: Server hosting the web application

Server-side Attacks

- Information Disclosure
- SSRF
- Directory Traversal
- File Upload
- Authentication
- Access Control
- SQL Injection
- Command Injection
- DDOS

… etc

Client Side Attacks

- CSRF
- XSS
- CORS
- ClickJacking
- Web Sockets

… etc

Outline

- Server-side Attacks and Defense
- Client-side Attacks and Defense
- Advanced/Complex Attacks
- Web Security Landscape
  - Anatomy of web attacks, OWASP top 10, CVE database and CVSS scores
  - Overall Defense and Best Practices

Enter the World of Web Security!